Worksheet 2 Solutions

MTH 203 (Multivariate Calculus)

Date: 25-08-2026

Q1. Find the interior, exterior and boundary of the sets

(a) $S_1 = \{x \in \mathbb{R}^n : ||x|| \leq 1\}$;

(b) $S_2 = \{x \in \mathbb{R}^n : ||x|| = 1\}$;

(c) $S_3 = \{x \in \mathbb{R}^n : x_i \text{ is rational}\}$.

Solution:

(a) Let

$$S_1 = \{x \in \mathbb{R}^n : \|x\| \leq 1\}.$$

We claim that

$$\boxed{S_1^\circ = \{x \in \mathbb{R}^n : \|x\| < 1\}.}$$

Indeed, let $x \in S_1$ with $\|x\| < 1$. Put

$$r = 1 - \|x\| > 0.$$

If $y \in B_r(x)$, then by the triangle inequality,

$$\|y\| \leq \|y - x\| + \|x\| < r + \|x\| = 1.$$

Hence

$$B_{1-\|x\|}(x) \subseteq S_1.$$

Therefore x is an interior point of S_1.

Now suppose that $\|x\| = 1$. For every $r > 0$, consider

$$y = \left(1 + \frac{r}{2}\right) x.$$

Then

$$\|y - x\| = \frac{r}{2} < r,$$

so $y \in B_r(x)$, while

$$\|y\| = 1 + \frac{r}{2} > 1.$$

Thus $y \notin S_1$. Hence no point with $\|x\| = 1$ is an interior point.

Therefore,

$$\boxed{S_1^\circ = \{x : \|x\| < 1\}.}$$

Consider any point $x \in S_1$ with $\|x\| = 1$. We show that every ball $B_r(x)$ contains points both from S_1 and outside S_1.
For $0 < r < 2$, let

$$y = \left(1 - \frac{r}{2}\right) x.$$

Then

$$\|y - x\| = \frac{r}{2}\|x\| = \frac{r}{2} < r,$$

so $y \in B_r(x)$, and
$$\|y\| = 1 - \frac{r}{2} < 1.$$
Thus $y \in S_1$.
On the other hand, let
$$z = \left(1 + \frac{r}{2}\right) x.$$
Then
$$\|z - x\| = \frac{r}{2} < r,$$
so $z \in B_r(x)$, while
$$\|z\| = 1 + \frac{r}{2} > 1.$$
Hence $z \notin S_1$.
Therefore every ball centered at x contains points both in S_1 and in S_1^c. Hence
$$\boxed{\{x : \|x\| = 1\} \subseteq \partial S_1.}$$
Moreover, let $x \in \mathbb{R}^n$ with $\|x\| > 1$. We claim that
$$B_{\frac{\|x\|-1}{2}}(x) \subseteq S_1^c.$$
Indeed, if
$$y \in B_{\frac{\|x\|-1}{2}}(x),$$
then, by the reverse triangle inequality,
$$\|y\| \geq \|x\| - \|y - x\| > \|x\| - \frac{\|x\| - 1}{2} = \frac{\|x\| + 1}{2} > 1.$$
Therefore $y \notin S_1$, and hence
$$\boxed{B_{\frac{\|x\|-1}{2}}(x) \subseteq S_1^c.}$$
Thus every point x with $\|x\| > 1$ is an exterior point of S_1.
Therefore,
$$\boxed{\begin{aligned} S_1^\circ &= \{x : \|x\| < 1\}, \\ S_1^e &= \{x : \|x\| > 1\}, \\ \partial S_1 &= \{x : \|x\| = 1\}. \end{aligned}}$$

(b) Let
$$S_2 = \{x \in \mathbb{R}^n : \|x\| = 1\}.$$
This is the unit sphere in $\mathbb{R}^n$.
We claim that
$$\boxed{S_2^\circ = \varnothing.}$$
Indeed, let $x \in S_2$, so that $\|x\| = 1$. For any $r > 0$, consider
$$y = \left(1 + \frac{\min\{r, 1\}}{2}\right) x.$$
Then
$$\|y - x\| = \frac{\min\{r, 1\}}{2} < r,$$
so $y \in B_r(x)$. But
$$\|y\| = 1 + \frac{\min\{r, 1\}}{2} > 1,$$
and therefore $y \notin S_2$.

Thus no open ball centered at a point of S_2 is contained in S_2. Hence

$$\boxed{S_2^\circ = \varnothing.}$$

It is easy to verify that $\partial S_2 = S_2$.
Finally, the exterior is $S_2^e = \mathbb{R}^n \setminus (\partial S_2 \cup (S_2)^\circ)$. Hence

$$\boxed{S_2^e = \{x \in \mathbb{R}^n : \|x\| \neq 1\}.}$$

Therefore,

$$\boxed{\begin{aligned} S_2^\circ &= \varnothing, \\ S_2^e &= \{x : \|x\| \neq 1\}, \\ \partial S_2 &= \{x : \|x\| = 1\}. \end{aligned}}$$

(c) Let

$$S_3 = \{x \in \mathbb{R}^n : x_i \text{ is rational for every } i\}.$$

We claim that

$$\boxed{S_3^\circ = \varnothing.}$$

Let $x \in S_3$ and let $r > 0$. We show that $B_r(x)$ contains a point which is not in S_3.
Since the irrational numbers are dense in $\mathbb{R}$, we can choose an irrational number a such that

$$|a - x_1| < r.$$

Define

$$y = (a, x_2, \ldots, x_n).$$

Then

$$\|y - x\|^2 = (|a - x_1|^2 + |x_2 - x_2|^2 + \cdots + |x_n - x_n|^2) = |a - x_1|^2 < r^2,$$

so

$$y \in B_r(x).$$

However, a is irrational, and therefore

$$y \notin S_3.$$

Thus no open ball centered at a point of S_3 is contained in S_3. Hence

$$\boxed{S_3^\circ = \varnothing.}$$

Next, consider any point $x \in \mathbb{R}^n$ and any ball $B_r(x)$ of radius r centered at x. Pick rational number $r_1, r_2, \cdots, r_n \in \mathbb{Q}$ such that $|r_i - x_i| < r/(\sqrt{n})$, then

$$\begin{aligned} (|r_1 - x_1|^2 + |r_2 - x_2|^2 + \cdots + |r_n - x_n|^2) &= r^2/n + r^2/n + \cdots + r^2/n \\ &= r^2 \end{aligned}$$

Thus, $||y - x|| < r$ where $y = (r_1, r_2, \cdots, r_n) \in S_3$. Similary pick $z = (s_1, x_2, \cdots, x_n)$ such that each s_1 is irrational and $||z - x|| < r$ which shows that $B_r(x)$ contains point $y \in S_3, z \notin S_3$. Thus

$$\boxed{\partial S_3 = \mathbb{R}^n.}$$

Finally,
$S_2^e = \mathbb{R}^n \setminus (\partial S_2 \cup (S_2)^\circ) = \mathbb{R}^n \setminus \mathbb{R}^n = \varnothing$. Therefore,

$$\boxed{\begin{aligned} S_3^\circ &= \varnothing, \\ S_3^e &= \varnothing, \\ \partial S_3 &= \mathbb{R}^n. \end{aligned}}$$

Q2. Prove that the union of any (even infinite) number of open sets is open. Prove that the intersection of two (and hence of finitely many) open sets is open. Give a counterexample for infinitely many open sets.

Proof: Let $A_i \subseteq \mathbb{R}^n, i \in I$ be a collection of open sets in $\mathbb{R}^n$. Consider $x \in \bigcup_{i\in I} A_i$, which means for some $j \in I$; $x \in A_j$. Therefore, there exists a $B_r(x) \subseteq A_j \subseteq \bigcup_{i\in I} A_i$ which concludes that $\bigcup_{i\in I} A_i$ is open set.
Let $A_1, A_2 \subseteq \mathbb{R}^n$ be a non-empty open sets. Consider, $x \in A_1 \cap A_2$ and choose $B_{r_1}(x) \subseteq A_1$ and $B_{r_2}(x) \subseteq A_2$. For $r = \min\{r_1, r_2\}$, we have

$$B_r(x) \subseteq B_{r_1}(x) \cap B_{r_2}(x) \subseteq A_1 \cap A_2.$$

which shows that $A_1 \cap A_2$ is open in $\mathbb{R}^n$. For finitely many open set, we can apply the principle of mathematical induction.
For $n \in \mathbb{N}$, consider $A_n = \left(\frac{-1}{n}, \frac{1}{n}\right)$. Prove that $\cap A_n = \{0\}$ which shows that arbitrary intersection of open set need not be open.

Q3. Let S be a subset of $\mathbb{R}^n$. Prove that $\text{int}(S)$ is open subset of $\mathbb{R}^n$.

Proof: Let $y \in \text{int}(S)$. By definition there exists a ball, $B_r(y) \subseteq S$. It is enough to prove that $B_{r/2}(y) \subseteq \text{int}(S)$. To see this, fix $z \in B_{r/2}(y)$ and $z' \in B_{r/2}(z)$, then

$$||z' - y|| \leq ||z' - z|| + ||z - y|| < r/2 + r/2 = r$$

In particular we have

$$B_{r/2}(z) \subseteq B_r(y) \subseteq S$$

which shows that z is an interior point and hence $B_{r/2}(y) \subseteq \text{int}(S)$.

Q4. Let S be the set of all points $(x, y) \in \mathbb{R}^n$ satisfying the given conditions. Make a sketch showing the set S and give a geometric argument to explain whether S is open, closed, both open and closed, or neither open nor closed.

(a) $y = x^2$;

(b) $1 < x^2 + y^2 \leq 2.$;

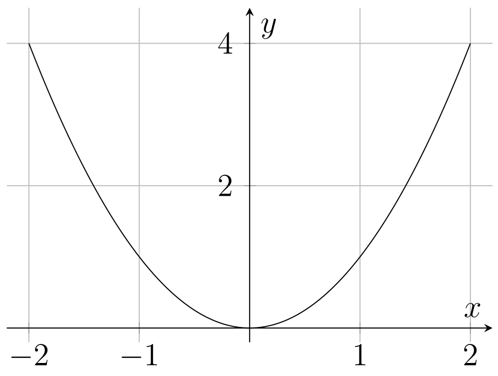
(c) $1 \leq x \leq 2, 3 \leq y \leq 4.$.

Solution:

(a)

$$S = \{(x, y) \in \mathbb{R}^2 : y = x^2\}.$$

This is the parabola $y = x^2$, including all the points on the parabola.



Not open: Let $z = (a, a^2) \in S$. Every open ball $B_r(z)$, no matter how small $r > 0$ is, contains points which are not on the parabola. For example, for sufficiently small $\varepsilon > 0$,

$$(a, a^2 + \varepsilon) \notin S,$$

although it can be made arbitrarily close to z.
Hence no point of S is an interior point. Therefore,

$$\boxed{S \text{ is not open.}}$$

Closed: Using the same argument above show that $S = \partial S$. Therefore, S is closed.

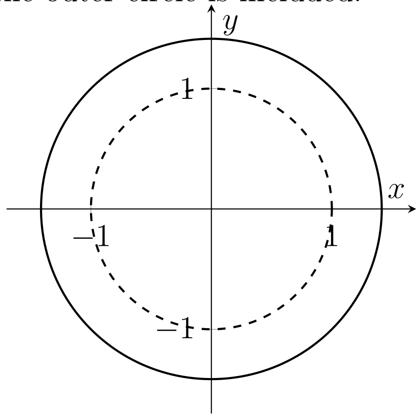
(b)

$$S = \{(x, y) \in \mathbb{R}^2 : 1 < x^2 + y^2 \leq 2\}.$$

Since $x^2 + y^2 = r^2$, this condition is equivalent to

$$1 < r \leq \sqrt{2}.$$

Thus S is the annulus between the circles $r = 1$ and $r = \sqrt{2}$, where the inner circle is excluded and the outer circle is included.



Not open: Take a point z on the outer circle $x^2 + y^2 = 2$. Then $z \in S$. However, every open ball centered at z contains points satisfying

$$x^2 + y^2 > 2,$$

which are not in S.
Therefore,

$$\boxed{S \text{ is not open.}}$$

Not closed: Consider any point z on the unit circle. Since every open ball $B_r(z)$ centered at z contain point $(x, y) \notin S$ with $x^2 + y^2 < 1$. This shows that z is the boundary point of S but $z \notin S$. Hence

$$\boxed{S \text{ is not closed.}}$$

Therefore,

$$\boxed{S \text{ is neither open nor closed.}}$$

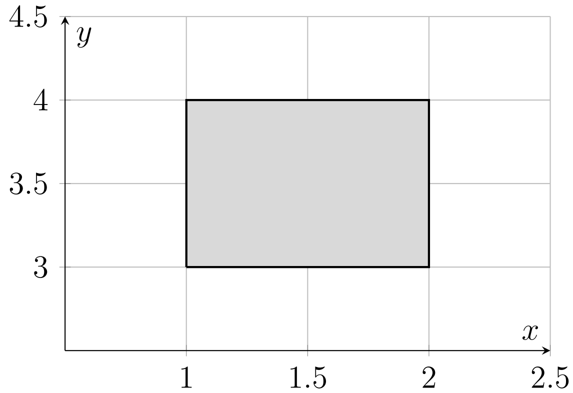
(c)

$$S = \{(x, y) \in \mathbb{R}^2 : 1 \leq x \leq 2,\ 3 \leq y \leq 4\}.$$

This is the rectangle

$$S = [1, 2] \times [3, 4],$$

with all four sides included.



Not open: Consider a boundary point, for example $(1, 3) \in S$. Every open ball centered at $(1, 3)$ contains points such as

$$(1 - \varepsilon, 3),$$

which lie outside S.
Therefore, S is not open:

$$\boxed{S \text{ is not open.}}$$

Closed: Choose any point on the boundary of rectangle, then any ball around this point contains point from the set S and outside S and hence $\partial S = \{(x, a) : 1 \leq x \leq 2, a = 3, 4\} \cup \{(b, y) : 3 \leq y \leq 4, b = 1, 2\} \subseteq S$. Hence,

$$\boxed{S \text{ is closed but not open.}}$$

Q5. Find the limit of f as $(x, y) \to (0, 0)$ or show that the limit does not exist (using polar coordinate)

(a) $f(x, y) = \cos\left(\frac{x^3 - y^3}{x^2 + y^2}\right)$;

(b) $f(x, y) = \tan^{-1}\left(\frac{|x| + |y|}{x^2 + y^2}\right)$;

(c) $f(x, y) = \frac{x^2 - y^2}{x^2 + y^2}$.

Solution:

(a) $\lim\limits_{(x,y)\to(0,0)} \cos\left(\frac{x^3 - y^3}{x^2 + y^2}\right) = \lim\limits_{r\to 0} \cos\left(\frac{r^3\cos^3(\theta) - r^3\sin^3(\theta)}{r^2\cos^2(\theta) + r^2\sin^2(\theta)}\right) = \lim\limits_{r\to 0} \cos\left(\frac{r(\cos^3(\theta) - \sin^3(\theta))}{1}\right) = 1$.

(b) $\lim\limits_{(x,y)\to(0,0)} \tan^{-1}\left(\frac{|x| + |y|}{x^2 + y^2}\right) = \lim\limits_{r\to 0} \tan^{-1}\left(\frac{|r\cos\theta| + |r\sin\theta|}{r^2}\right)$. If $r \to 0^+$, then

$\lim\limits_{r\to 0^+} \tan^{-1}\left(\frac{|r\cos\theta| + |r\sin\theta|}{r^2}\right) = \tan^{-1}\left(\frac{|\cos\theta| + |\sin\theta|}{r}\right) = \pi/2$; if $r \to 0^-$ then

$$\lim_{r\to 0^-} \tan^{-1}\left(\frac{|r\cos\theta|+|r\sin\theta|}{r^2}\right) = \tan^{-1}\left(\frac{|\cos\theta|+|\sin\theta|}{-r}\right) = \pi/2.$$

(c) $\lim\limits_{(x,y)\to(0,0)} \frac{x^2-y^2}{x^2+y^2} = \lim\limits_{r\to 0} \frac{r^2\cos^2\theta - r^2\sin^2\theta}{r^2} = \lim\limits_{r\to 0}(\cos^2\theta - \sin^2\theta) = \lim\limits_{r\to 0}\cos 2\theta$ which ranges between -1 and 1 depending on θ. The limit does not exists.

Q6. Determine the largest set on which the given functions are continuous

(a) $f(x,y) = \frac{1}{x^2-y}$;

(b) $f(x,y,z) = \sqrt{x+y+z}$;

(c) $f(x,y,z) = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$

(d) $f(x,y,z) = \frac{1}{|xy|+|z|}$.

Solution:
(a) $S_1 = \{(x,y) : y \neq x^2\}$.

(b) $S_2 = \{(x,y,z) : x+y+z \geq 0\}$.

(c) Homework.

(d) All $(x,y,z) \in \mathbb{R}^3$ except $(0,y,0)$ or $(x,0,0)$.

Q7. Define $f(0,0)$ in a way that extends $f(x,y) = xy\frac{x^2-y^2}{x^2+y^2}$ to be continuous at the origin.

Solution: $\lim\limits_{(x,y)\to(0,0)} xy\frac{x^2-y^2}{x^2+y^2} = \lim\limits_{r\to 0}(r\cos\theta)(r\sin\theta)\frac{r^2\cos^2\theta - r^2\sin^2\theta}{r^2} = \lim\limits_{r\to 0} r^2(\cos^2\theta - \sin^2\theta) = 0$. Define $f(0,0) = 0$ such that f is continuous at $(0,0)$.

Q8. In each example below determine the limit

(a) $\lim_{(x,y)\to(0,0)} e^y \frac{\sin x}{x}$;

(b) $\lim_{(x,y)\to(\pi/2,0)} \frac{\cos y+1}{y-\sin x}$;

(c) $\lim_{(x,y)\to(1,1)} \ln|1+x^2y^2|$;

(d) $\lim_{(x,y)\to(0,\ln 2)} e^{x-y}$.

Proof:
(a) $\lim_{(x,y)\to(0,0)} e^y \frac{\sin x}{x} = (\lim_{(x,y)\to(0,0)} e^y)(\lim_{(x,y)\to(0,0)} \frac{\sin x}{x}) = 1$.

(b) $\lim_{(x,y)\to(\pi/2,0)} \frac{\cos y+1}{y-\sin x} = \frac{\cos 0+1}{0-\sin\pi/2} = -2$.

(c) $\lim_{(x,y)\to(1,1)} \ln|1+x^2y^2| = \ln|1+1| = \ln 2$.

(d) $\lim_{(x,y)\to(0,\ln 2)} e^{x-y} = e^{0-\ln 2} = e^{-\ln 2} = 1/2$

Bonus Questions (No marks)

Q1. Let $f(x,y) = 0$ if $y \leq 0$ or if $y \geq x^2$ and let $f(x,y) = 1$ if $0 < y < x^2$. Show that $f(x,y) \to 0$ as $(x,y) \to (0,0)$ along any straight line through the origin.

Solution: Consider an arbitrary straight line through $(0,0)$:

$$y = mx, \qquad m \in \mathbb{R}.$$

Case 1: If $m < 0$.
For $x \geq 0$, $mx \leq 0$ so $f(x,mx) = 0$.
For $x < 0$, $mx > 0$, but

$$mx > x^2$$

for sufficiently small $x < 0$. Hence $f(x,mx) = 0$ near the origin.

Case 2: $m = 0$, then $f(x,mx) = f(x,0) = 0$.

Case 3: $m > 0$.
For $x \leq 0$, $mx \leq 0$ so $f(x,mx) = 0$.
For $x > 0$, since $m > 0$, for sufficiently small x we have $x < m$ and hence $x^2 < mx$.
Hence $f(x,mx) = 0$ near the origin. Thus,

$$\lim_{(x,y)\to(0,0):y=mx} f(x,y) = (0,0)$$

Q2. In the previous problem (1), find a curve through the origin along which (except at the origin) $f(x,y)$ has the constant-value 1. Is f continuous at the origin?

Solution: Consider the curve C given by the equation $y = x^2/2$, then $x \neq 0$ we have $0 < x^2/2 < x^2$ and hence $f(x,y) = 1$. Along this curve, $f(x,y) \to 1$ as $(x,y) \to (0,0)$.
On the other hand, along any straight line we have,

$$\lim_{(x,y)\to(0,0):y=mx} f(x,y) = (0,0)$$

Thus f is not continuous at the origin.